

Evaporation

UNIT OPERATIONS LAB

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SANDRA HERNANDEZ

TIANNA MITCHELL

ANNE SERBAN

MAURICE SMITH



Introduction

Experimental objective: determine the capacity and economy at 3 different vacuum pressures

- Become familiar with single-effect evaporation process
- Operate an evaporator at constant concentration
- Understand effect vacuum pressure has on boiling point of solutions.

Theory

Two principal measures of performance: capacity and economy

Capacity = kg of water vaporized / hr

Economy = kg vaporized / kg of steam fed

If evaporator feed is at boiling temperature, all heat transferred through heating surface can be evaporated and the capacity is proportional to heat transfer rate

$$q = UA\Delta T$$

Where:

U is the overall heat transfer coefficient

A is the heat transfer surface

ΔT is the overall temperature drop

Theory

Enthalpy balance for steam side:

$$q_s = \dot{m}_s(H_s - H_c) = \dot{m}_s\lambda_s$$

Where:

q_s = rate of heat transfer through heating surface from steam

H_s is the specific enthalpy of steam

H_c is the specific enthalpy of condensate

λ_s is the latent heat of condensation of steam

$\dot{m}_s$ is the rate of flow of steam

Theory

Enthalpy balance for liquor side:

$$q = (\dot{m}_f - \dot{m})H_v - \dot{m}_f H_f + \dot{m}H$$

Where:

q is the rate of heat transfer from heating surface to liquid

H_v is the specific enthalpy of vapor

H_f is the specific enthalpy of thin liquor

H is the enthalpy for thick liquor

$\dot{m}_f$ and $\dot{m}$ is the rate of steam flow of the thin liquor or feed and the rate of steam flow of the thick liquor respectively

Apparatus

Table 1: Apparatus Parts and Description

No.	Instrument	Description
1	Flow Valve	Controls flow rate of water used to condense vapor
2	Pressure Valve	Controls vacuum pressure applied to system
3	Evaporation Vessel	Contains solution to be evaporated
4	Erlenmeyer Flasks	Recovers condensed steam
5	Heat Exchanger	Evaporates solution
6	Condenser	Condenses water vapor to liquid water.
7	Digital Temp. Reader	Displays temperature readings for 8 thermocouples throughout system
8	Temperature Gauge	Displays temperature of water before and after condenser.
9	Rotameter	Displays volumetric flow rate of the cooling water
10	Steam Trap	
11	Vacuum Pump	Adjust to obtain the desired pressure in the separator
12	Water Makeup Tank	Contains pure water to flow and maintain constant level
13	Steam Valve [Not Pictured]	Controls steam input to system, located behind the apparatus
14	Evaporator Pump Power Switch	Turns on and off the evaporator pump

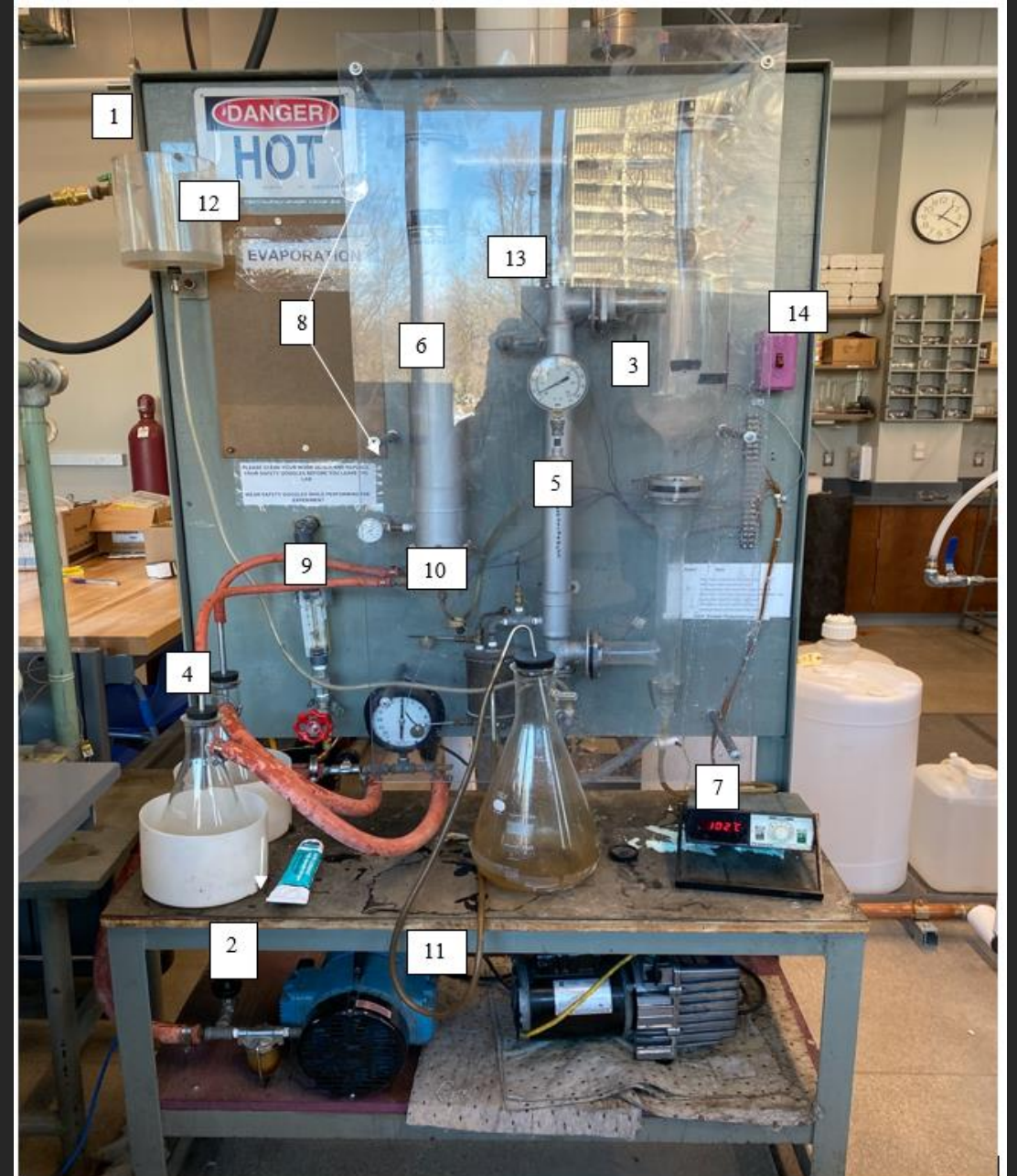


Figure 1: Labeled Evaporation Apparatus

Materials & Supplies

Materials & Supplies	Purpose
Graduated Cylinder	Collect steam condensate
Distilled Water	Maintain steady state level from makeup tank and make solution
Cooling Water	Condense remaining steam to steam trap
Stopwatch	Measure volumetric flow rate
Steam	Supplied to heat exchanger
Sugar	To prepare 10 wt% sugar water solution
Scoopula	Used to transfer sugar to make solution
Scale	Measure weight of sugar
Funnel	To add solution to the separator
Heat-resistance gloves	To protect hands from hot liquids
Vacuum grease	To ensure that there is a good seal throughout system

Procedure

Prepare 10 wt% sugar solution and ensure the separator is completely drained.

Fill the separator to the level line with the sugar solution.

Add distilled water in the makeup reservoir

Start the steam flow.

Start the cooling water to the steam trap and collect the condensate in a graduated cylinder.

Turn on the vacuum pump.

- Adjust bypass valve to a desired pressure (reads in inches of mercury).
- Adjust the flow of the vacuum until desired pressure is reached.

Collect vapor product being conscious that the concentration of the solution will not change.

- Done by adjusting make-up water flow rate.

Procedure

Allow the system to reach steady state (based on the temperature readings).

Record the level of the steam condensate and the collected vapor.

Wait 15 minutes; record the time and the levels again.

Calculate the flowrates of the two streams.

Record the inlet and outlet temperature and the flowrate of cooling water.

Determine how much steam has condensed in the exchanger.

Change the pump to a different vacuum pressure for two additional trials.

Experimental Challenges

- Making enough 10 wt.% solution to reach the fill line.
- Filling the separator vessel with the solution.
- Keeping the level in the separator constant.
- Keeping the vacuum pressure constant.
- Makeup water flowrate extremely sensitive.

Safety

Exposed wiring near separator vessel → be careful when filling it

Vacuum pressure could cause cracked vessels to implode.

Steam is at high temperature and pressure.



QUESTIONS?